

# End-to-End Machine Learning on Retail Transaction Data

A Theory-and-Results Report on the Sample Superstore Dataset

Debarshi Chakraborty

September 1, 2026

Companion codebase: `EY Training/ml_superstore/`  
Data: `EY Training/data/Sample - Superstore.csv` (9,994 order lines, 2014–2017,  
USA)

## Abstract

This report documents an end-to-end machine learning study built on the Sample Superstore retail dataset. It follows the same five stages as the accompanying Jupyter notebooks and source package (`EY Training/ml_superstore/`): data ingestion and quality assessment, exploratory data analysis, feature engineering, time-series forecasting of monthly sales, and supervised learning for line-level profit prediction and loss detection. Unlike the notebooks, whose purpose is interactive exploration, this report is written to stand alone: every technique used is first introduced from first principles — the statistical or algorithmic theory it rests on — and only then applied to the data, with the resulting figures, tables and their business interpretation presented immediately afterwards. The headline findings are that (i) order discounting beyond roughly 30% is the dominant driver of loss-making order lines, far outweighing category, region or shipping mode; (ii) an additive Holt–Winters exponential smoothing model forecasts monthly sales with an out-of-sample MAPE of approximately 23%, competitive with a seasonal ARIMA model and clearly better than a seasonal-naïve baseline; and (iii) a random-forest classifier trained on leakage-safe features identifies loss-making order lines with a PR-AUC of 0.96 on a strictly future (2017) hold-out year, giving a concrete, auditable basis for a discount governance policy.

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# Chapter 1

## Introduction

### 1.1 Objective

Retail order-line data — one row per product sold within an order — sits at the intersection of three classical machine-learning problem types: descriptive analytics (what happened), predictive analytics (what will happen), and diagnostic/prescriptive analytics (why it happened, and what to do about it). This report uses the public *Sample Superstore* dataset to walk through all three, mirroring a realistic analytics engagement:

1. **Data quality and exploratory analysis** — establish that the data is trustworthy and characterise its statistical structure.
2. **Feature engineering** — turn raw transactional fields into model-ready predictors, with explicit attention to information leakage.
3. **Forecasting** — model and project the aggregate monthly sales time series.
4. **Supervised learning** — model two line-level targets, *profit* (regression) and *is a line loss-making* (classification), and interpret what drives them.

Each stage in this report is structured identically: a **Theory** section derives or states the relevant statistical/ML machinery from first principles, followed by a **Method** section describing how it was applied to this dataset, then **Results** with the produced figures and tables, and finally an **Interpretation** section connecting the numbers back to a business reading. This mirrors, chapter for chapter, the five numbered notebooks in `EY_Training/ml_superstore/notebooks/` and the reusable modules in `src/`.

### 1.2 Dataset

The dataset contains  $n = 9,994$  order lines spanning 4 years (2014-01-03 to 2017-12-30), 793 unique customers and 1,862 unique products, organised into 5,009 distinct orders across the United States. Table 1.1 summarises the raw schema.

**Table 1.1:** Raw schema of the Sample Superstore CSV.

| Field                                            | Type          | Description                           |
|--------------------------------------------------|---------------|---------------------------------------|
| Row ID, Order ID                                 | id            | Line and order identifiers            |
| Order Date, Ship Date                            | date          | Order placement and shipment dates    |
| Ship Mode                                        | categorical   | Standard / Second / First Class, Same |
| Customer ID, Customer Name                       | id/text       | Buyer identity                        |
| Segment                                          | categorical   | Consumer / Corporate / Home Office    |
| Country, City, State, Postal Code, Region        | geography     | U.S. location fields                  |
| Product ID, Category, Sub-Category, Product Name | catalogue     | 3-level product hierarchy             |
| Sales                                            | numeric (\$)  | Line revenue                          |
| Quantity                                         | numeric       | Units sold                            |
| Discount                                         | numeric [0,1) | Fractional discount applied           |
| Profit                                           | numeric (\$)  | Line profit (can be negative)         |

### 1.3 Software and reproducibility

All analysis is implemented in Python 3.10 using `pandas`, `numpy`, `scikit-learn`, `statsmodels`, `matplotlib/seaborn`. Code lives in `EY_Training/ml_superstore/src/` as importable modules (`data_loader`, `features`, `forecasting`, `modeling`, `pipeline`), exercised interactively by six Jupyter notebooks and, for one-shot reproduction, by `python -m src.pipeline`. Every figure reproduced in this report was generated by that codebase and saved under `outputs/figures/`; every number quoted in the text is read directly from the corresponding CSV in `outputs/tables/` or the machine-written `outputs/REPORT.md`. No figure or statistic in this document was computed by hand.

### 1.4 Notation

Throughout,  $y_i$  denotes a target value for observation  $i$ ,  $\hat{y}_i$  its model prediction,  $\mathbf{x}_i \in \mathbb{R}^p$  its feature vector, and  $n$  the sample size of the set under discussion (which may be a train, test, or full-data set, made explicit in context). For the monthly sales series we write  $y_t$ ,  $t = 1, \dots, T$ , for the value in month  $t$ .  $\mathbb{E}[\cdot]$ ,  $\text{Var}(\cdot)$  and  $\text{Cov}(\cdot, \cdot)$  denote expectation, variance and covariance under whatever probabilistic model is being assumed at that point (population model for theory, empirical/plug-in estimator when applied to data).

## Chapter 2

# Data Quality and Exploratory Analysis

## 2.1 Theory

### 2.1.1 Data quality auditing

Before any modelling, three questions must be answered about a tabular dataset  $D = \{(\mathbf{x}_i)\}_{i=1}^n$ : (i) is every field typed and parsed correctly, (ii) is any value missing, and (iii) is any row duplicated. Formally, for a column  $X$  we report the *missingness rate*  $\hat{\pi}_X = \frac{1}{n} \sum_{i=1}^n \mathbb{I}[x_i \text{ is NA}]$  and *cardinality*  $|\{x_i\}|$ . A duplicate row under a primary key  $k$  is any pair  $i \neq j$  with  $k_i = k_j$ ; if such pairs exist, the key does not uniquely identify a record and downstream joins or grouped aggregates will silently double-count. These checks are cheap and guard against exactly the class of error (a re-exported CSV with duplicated header rows, an encoding mismatch producing NA in a text field) that otherwise corrupts every later stage silently.

### 2.1.2 Descriptive statistics and distribution shape

For a numeric column  $X$  with observed values  $x_1, \dots, x_n$ , the sample mean  $\bar{x} = \frac{1}{n} \sum_i x_i$  and variance  $s^2 = \frac{1}{n-1} \sum_i (x_i - \bar{x})^2$  summarise location and spread, but retail money variables are rarely well described by them alone because they are *right-skewed*: many small transactions and a long tail of large ones. Skewness is measured by the third standardised moment,

$$\gamma_1 = \frac{\frac{1}{n} \sum_i (x_i - \bar{x})^3}{\left(\frac{1}{n} \sum_i (x_i - \bar{x})^2\right)^{3/2}}, \quad (2.1)$$

with  $\gamma_1 > 0$  indicating a right tail. A common remedy that stabilises both skew and the variance-vs-mean relationship (heteroscedasticity) seen in multiplicative processes like sales is the  $\log(1+x)$  transform, used throughout this project on **Sales** (and, with a sign-preserving variant, on **Profit**) purely for *visualisation*; models are always trained on the original scale unless stated otherwise, since tree ensembles are invariant to monotone transforms and linear models' coefficients are more directly interpretable untransformed here.

### 2.1.3 Correlation

For two numeric variables  $X, Y$ , the Pearson correlation coefficient

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}} \quad (2.2)$$



measures *linear* association,  $\rho_{XY} \in [-1, 1]$ . It is insensitive to non-linear relationships (e.g. a U-shape can have  $\rho \approx 0$  despite strong dependence), which is precisely why Section 2.3.4 supplements the correlation matrix with a binned conditional-mean plot: discount and profit turn out to have a threshold-like, not linear, relationship.

### 2.1.4 Seasonal-trend decomposition (STL)

A time series  $y_t$  is classically modelled as the additive sum of a slowly-varying trend  $T_t$ , a periodic seasonal component  $S_t$  (period  $L$ , here  $L = 12$  months), and a remainder  $R_t$ :

$$y_t = T_t + S_t + R_t. \quad (2.3)$$

**STL** (Seasonal-Trend decomposition using LOESS, Cleveland et al., 1990) estimates  $T_t$  and  $S_t$  by alternating two local regression (LOESS) smooths: it removes a running estimate of  $T_t$  from  $y_t$  to expose the seasonal cycle, smooths within each seasonal sub-series (all Januaries, all Februaries, ...) to obtain a smoothly time-varying  $S_t$ , then smooths  $y_t - S_t$  to update  $T_t$ ; this is iterated to convergence. Unlike a single global seasonal index, STL lets the seasonal shape drift slowly over the four years of data, and unlike classical decomposition it is robust to outliers when the **robust** option down-weights large residuals between iterations. We use STL purely as a diagnostic here; the forecasting models of ?? are fitted directly on  $y_t$ , not on the decomposed components.

## 2.2 Method

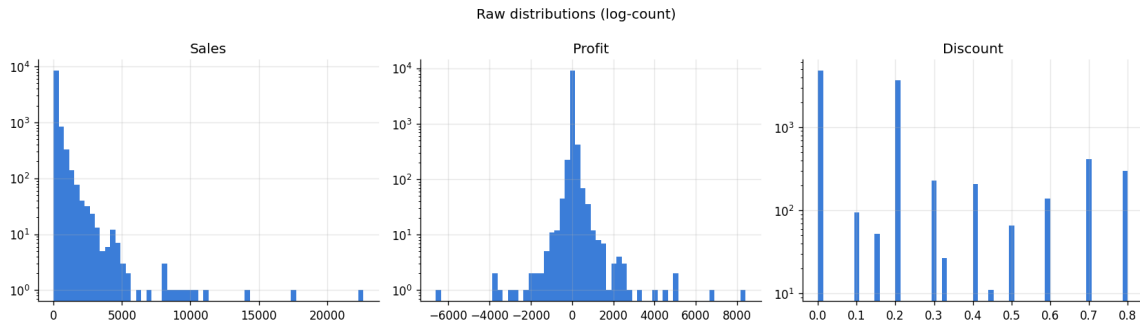
The raw CSV is read with `src.data_loader.load_raw` (encoding auto-detection across UTF-8/Latin-1/CP1252) and typed with `clean`, which parses `Order Date`/`Ship Date`, strips whitespace from every text column, drops exact and Row ID duplicates, and derives `Ship Days` = `Ship Date` − `Order Date`. The quality audit, univariate/bivariate plots and STL decomposition are produced by notebooks `01_data_overview.ipynb` and `02_edu.ipynb`.

## 2.3 Results

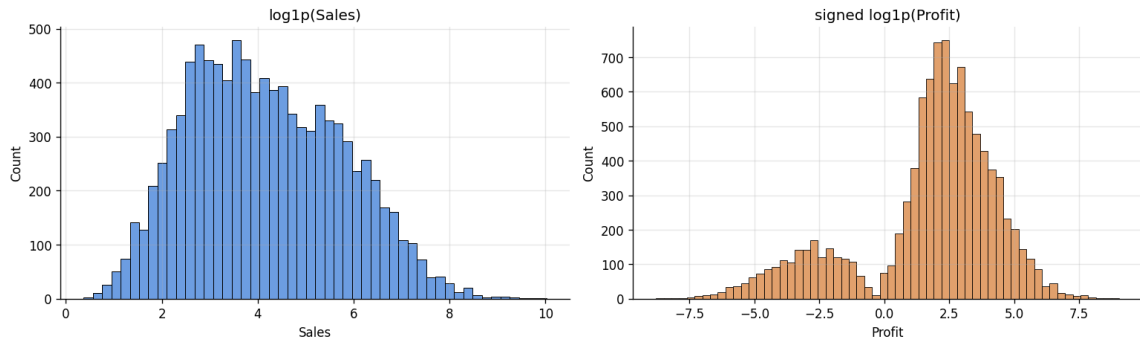
### 2.3.1 Data quality

The audit (`outputs/tables/01_data_quality.csv`) finds **zero missing values in any of the 21 raw columns**, no fully duplicated rows, and no duplicate Row IDs after cleaning — the data is analysis-ready without imputation. `Order Date` spans 2014-01-03 to 2017-12-30 (1,237 distinct dates); `Ship Days` is always non-negative, i.e. no order ships before it is placed.

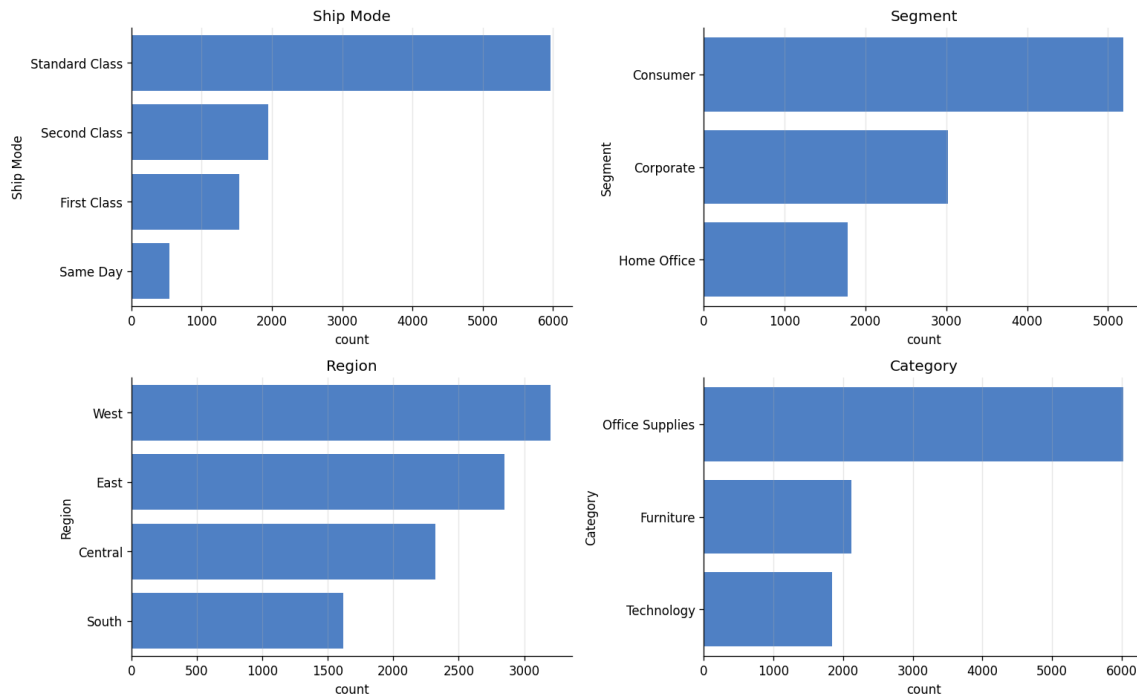
### 2.3.2 Distributions



**Figure 2.1:** Histograms (log-count  $y$ -axis) of **Sales**, **Profit** and **Discount** on the raw scale, showing pronounced right skew in **Sales** and a bimodal, discount-tier structure in **Discount**.



**Figure 2.2:** Left:  $\log(1 + \text{Sales})$  recovers an approximately unimodal, near-symmetric shape (Eq. (2.1) in the raw scale is large and positive; after the transform it is close to zero). Right:  $\text{signed } \log(1 + |\text{Profit}|)$ , showing the loss-making mass as a distinct negative lobe rather than a long left tail.



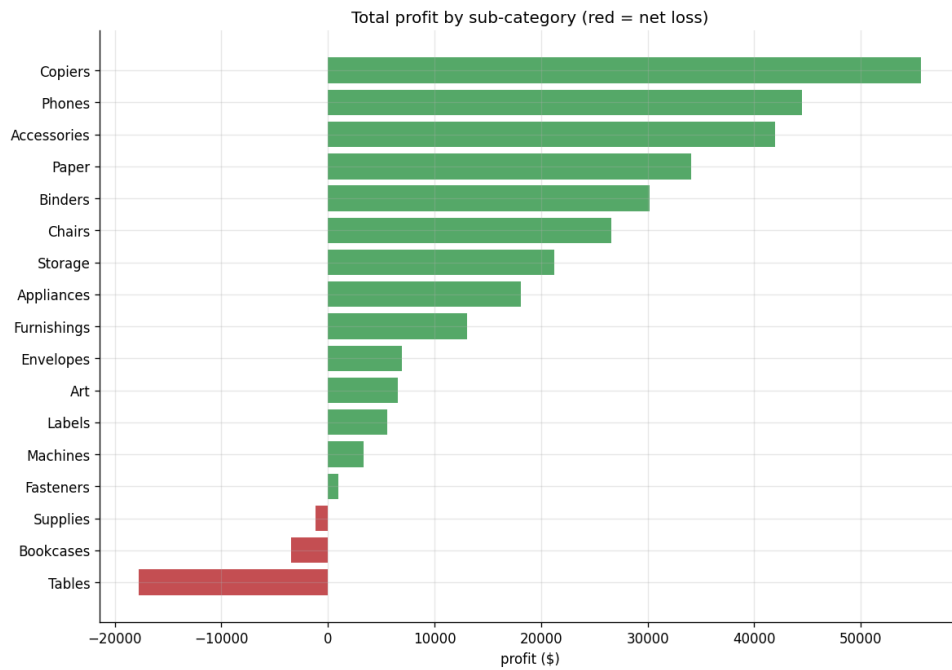
**Figure 2.3:** Order-line counts by Ship Mode, Segment, Region and Category. Standard Class and the Consumer segment dominate; the four regions are comparably sized, with West largest.

### 2.3.3 Profit by product hierarchy

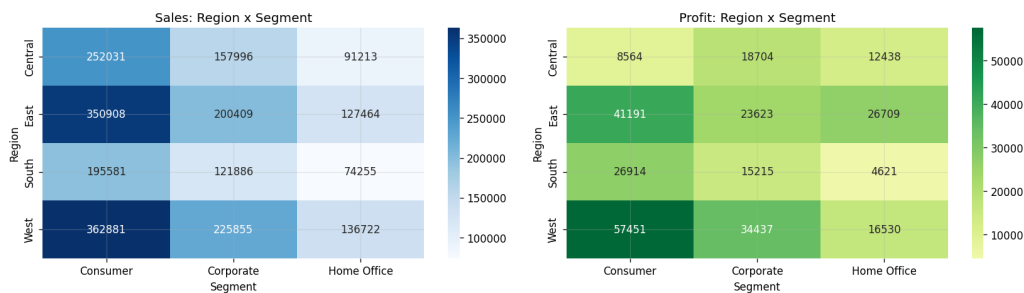
Table 2.1 (full detail in `outputs/tables/02_by_subcategory.csv`) aggregates profit by sub-category. **Tables** lose \$17,725 overall despite \$207k of sales, and **Bookcases** lose \$3,473 — both at average discounts above 21%, well above the store-wide mean. **Supplies** is also marginally loss-making. Every other sub-category is net profitable, led by Copiers (37% margin) and Phones (\$330k sales, 13% margin).

**Table 2.1:** Selected sub-categories by total profit (ascending). Full table has all 17 sub-categories.

| Category                               | Sub-Category | Sales (\$) | Profit (\$) | Avg. Discount | Margin       |
|----------------------------------------|--------------|------------|-------------|---------------|--------------|
| Furniture                              | Tables       | 206,966    | −17,725     | 26.1%         | −8.6%        |
| Furniture                              | Bookcases    | 114,880    | −3,473      | 21.1%         | −3.0%        |
| Office Supplies                        | Supplies     | 46,674     | −1,189      | 7.7%          | −2.5%        |
| ... (14 profitable sub-categories) ... |              |            |             |               |              |
| Technology                             | Phones       | 330,007    | 44,516      | 15.5%         | 13.5%        |
| Technology                             | Copiers      | 149,528    | 55,618      | 16.2%         | <b>37.2%</b> |

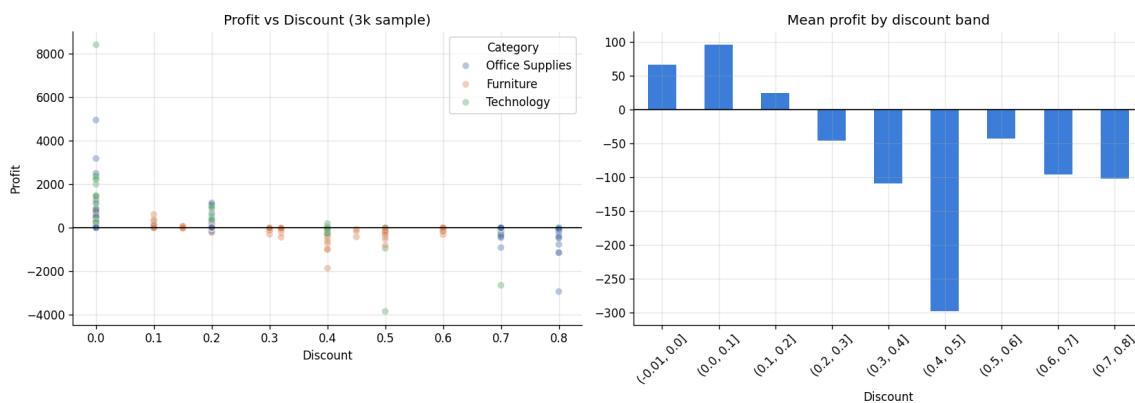


**Figure 2.4:** Total profit by sub-category; red bars are net loss-makers.



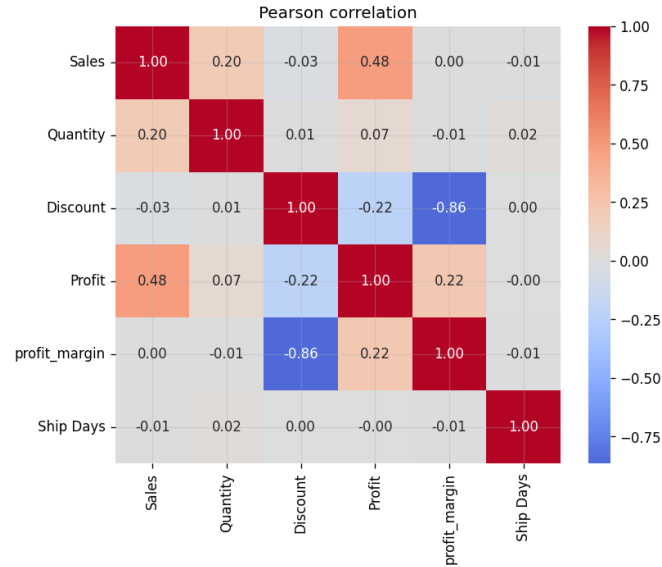
**Figure 2.5:** Sales (left) and profit (right) cross-tabulated by Region and Segment. Sales are fairly even across cells; profit is not — the Central/Consumer cell is comparatively weak.

### 2.3.4 Discount versus profit



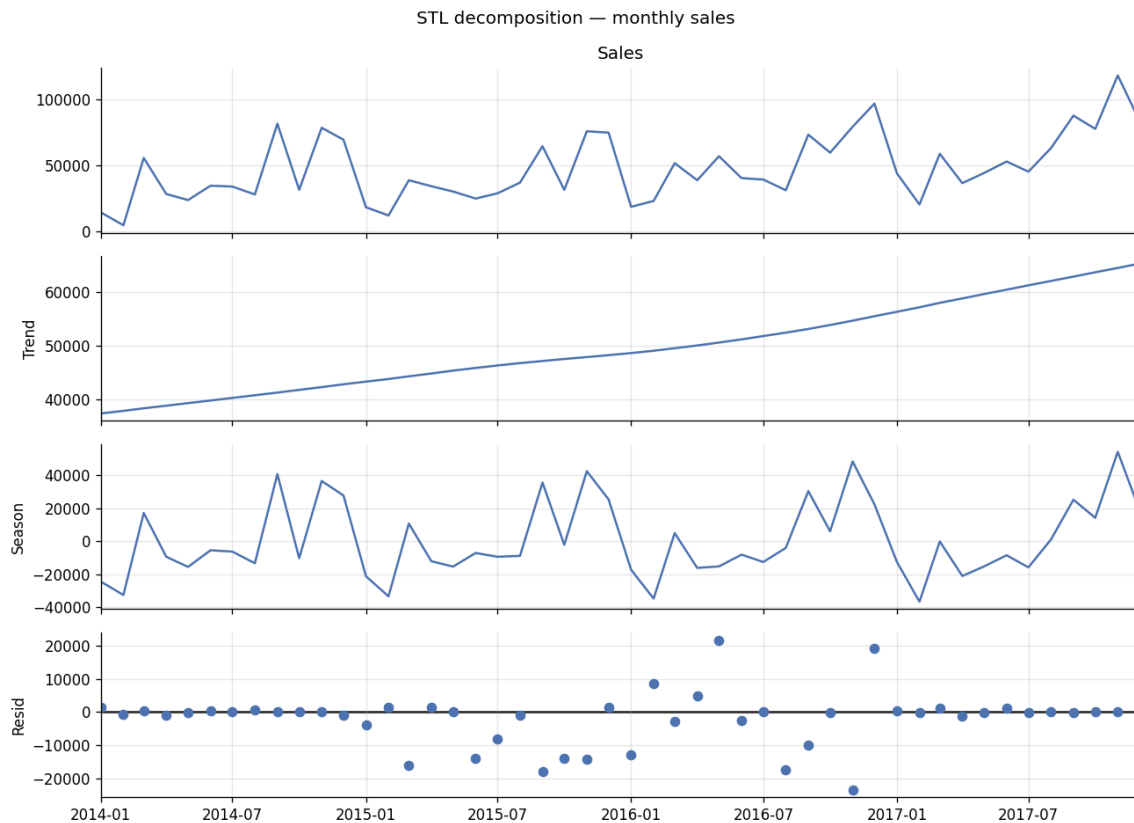
**Figure 2.6:** Left: line-level profit vs. discount (3,000-row sample), coloured by category — profit is scattered near zero at low discount and swings sharply negative as discount grows. Right: mean profit by discount band, crossing zero between the 10–20% and 20–30% bands and falling steeply beyond 30%.

Quantitatively, the loss rate ( $\Pr(\text{Profit} < 0)$ ) is **exactly 0%** for undiscounted lines and **96.8%** for lines discounted 30% or more, against an overall loss rate of 18.7% across all 9,994 lines — discount depth is close to a deterministic switch, not a mild risk factor. Because the relationship is a threshold effect rather than linear, the raw Pearson correlation between **Discount** and **Profit** (visible in Fig. 2.7) understates its practical importance; this motivates using the categorical `discount_bucket` feature (rather than relying on a linear term alone) in ????

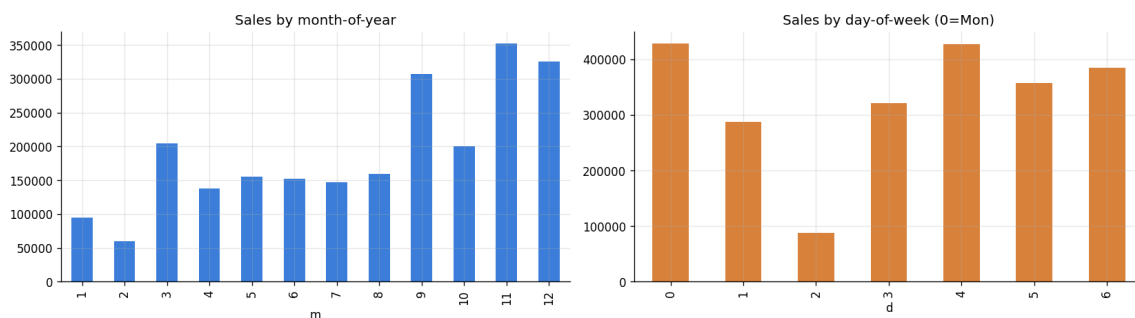


**Figure 2.7:** Pearson correlation (Eq. (2.2)) among the core numeric fields. **Sales–Profit:**  $\rho \approx 0.48$ ; **Discount–Profit:**  $\rho \approx -0.22$  (negative but, per Fig. 2.6, understating the non-linear effect); **Quantity** is nearly uncorrelated with profit.

### 2.3.5 Seasonality



**Figure 2.8:** STL decomposition (Eq. (2.3), period 12) of monthly sales: a clear upward trend across the four years and a recurring within-year seasonal pattern that peaks late in the year, leaving a comparatively small, structure-free remainder.



**Figure 2.9:** Sales aggregated by month-of-year (left) and day-of-week (right, 0 = Monday). November–December is by far the strongest period; day-of-week shows only mild variation.

### 2.3.6 Customers and products

The top-10 customers by sales (Table 2.2) generate between \$12k and \$25k each, but profitability is uneven within that list — *Sean Miller* is the single largest customer by revenue (\$25,043) yet is net loss-making (−\$1,981), while *Tamara Chand* generates comparable revenue at a much healthier margin. The same pattern repeats at product level: the *Cisco TelePresence System EX90* and two *GBC* binding systems are top-10 by revenue but net loss-making, again tracing back to discount depth on high-ticket Technology and Furniture items.

**Table 2.2:** Top 5 customers by total sales.

| Customer      | Sales (\$) | Profit (\$) | Orders |
|---------------|------------|-------------|--------|
| Sean Miller   | 25,043     | −1,981      | 5      |
| Tamara Chand  | 19,052     | 8,981       | 5      |
| Raymond Buch  | 15,117     | 6,976       | 6      |
| Tom Ashbrook  | 14,596     | 4,704       | 4      |
| Adrian Barton | 14,474     | 5,445       | 10     |

## 2.4 Interpretation

Three findings from this chapter shape every subsequent stage of the report. First, the data requires *no* missing-value or duplicate handling, so all modelling error downstream is genuinely about the signal, not data-cleaning artefacts. Second, **discount depth, not category or region, is the primary lever on profitability** — a threshold effect around 30% discount that is far stronger than any single-variable linear correlation suggests, which is why `discount_bucket` (??) and the corresponding permutation-importance results (??) recur as the dominant signal throughout the rest of the report. Third, the pronounced trend and December seasonality in Fig. 2.8 is exactly the structure a seasonal model must capture, and directly motivates the choice of Holt-Winters and SARIMA — both explicitly seasonal — over simpler alternatives in ??.